

Yun-Tang Chang

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Work Experience

Software Engineer (Backend Intern)

Sep 2023 - Jun 2025

NCKU Office of International Affairs (OIA)

- Built the backend for an agentic assistant serving 500+ students, with a custom MCP server for document retrieval.
- Optimized frequent-query handling with FAISS-based semantic caching, improving end-to-end throughput by 3×.
- Designed Redis-coordinated asynchronous worker pipelines to scale concurrent agent workflows.

Research Experience

Research Intern

Jun 2026 - Present

Department of ECE, Computer Engineering Group, University of Toronto

- Developing autoscaling strategies for I/O-sensitive stateful operators in the Flink Kubernetes Operator autoscaler.
- Modeling CPU-memory tradeoffs of stateful operators to guide autoscaling policies under constrained cluster resources.

Undergraduate Researcher

Feb 2026 - Jun 2026

Secure System Lab, National Taiwan University

- Integrated vsock with eBPF sockmap to delegate guest networking in SEV-SNP CVMs, reducing networking TCB by 10×.
- Optimized the vsock networking path, improving throughput by **21.8%** and HTTPS serving by **9.2%** over virtio-net.
- Profiled kernel-level vsock behavior with eBPF and perf to trace and eliminate performance bottlenecks.

Projects & Open Source

DuckDB

- Contributed upstream to DuckDB, a high-performance analytical database with **39.8K** GitHub stars.
- Optimized filter propagation across **4 query-optimizer rules** to prune additional data blocks during table scans.

duckdb-query-condition-cache

- Implemented a predicate cache in DuckDB's filter-pruning path to accelerate repeated dashboard queries.
- Sped up queries by up to **13.8×** on the HDFS_v2 log dataset by skipping decompression of non-qualifying data blocks.

duckdb-table-inspector

- Built DuckDB storage observability tooling to analyze database file usage and table-level compression efficiency.
- Reached 16K+ DuckDB community downloads and helped investigate free-block inflation and index-reclaiming issues.

Education

National Cheng Kung University (NCKU)

Taiwan

Bachelor of Science in Computer Science and Information Engineering

Sep. 2022 - Jun. 2026

- **Academics:** 4.19/4.3 GPA

Awards & Hackathons

2025 AWS Generative AI Hackathon – Taipei, Taiwan

Apr 2025

- Designed a retail analytics pipeline using Jetson devices, AWS Kinesis, and SageMaker for real-time heatmap inference.
- Integrated Lambda, S3, and API Gateway to support edge-to-cloud data streaming and analytics delivery.

Taipei Hackathon 2024, 3rd Place – Taipei, Taiwan

Sep 2024

- Engineered **ParkFlow**, a real-time parking and navigation app integrated into Taipei's **400,000-user** city platform.
- Reduced lookup latency by **53%** through geospatial search and database-query optimization.
- Implemented the system with Python, Vue 3, Flutter, and geospatial indexing over the Taipei Parking API.

Skills

Languages: C, C++, Python, Go, Rust

Systems / Data / Infra: Linux, eBPF, Redis, PostgreSQL, DuckDB, Apache Flink, Docker, Kubernetes, GitHub Actions

AI / Cloud: FAISS, AWS Kinesis, SageMaker, Lambda, S3